

FreshFlow Beverages Pvt. Ltd.

FreshFlow Bottling Plant - Pune Production Facility
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STANDARD OPERATING PROCEDURE

Shutdown Procedure for Water Treatment Line 50

Department: Water Treatment

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Prepared by: Engineering / QA Department
Approved by: Plant Manager

Document Control Information

Parameter	Value
SOP Number	SOP-050
Title	Shutdown Procedure for Water Treatment Line 50
Department	Water Treatment
Plant	FreshFlow Bottling Plant - Pune Production Facility
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1. Purpose

Produce water meeting the FreshFlow Beverages water quality specification (TDS <50 ppm, pH 6.5-7.5, microbiologically safe) for use in product production and CIP.

2. Scope

Water Treatment operators and Maintenance team.

3. Prerequisites and Requirements

3.1 Training Requirements

- RO system operation training
- Chemical dosing training

3.2 Tools and Materials Required

- Calibrated instruments as listed in the plant calibration schedule.
- Appropriate PPE as specified in Section 4.
- Completed permit-to-work (if required for maintenance activities).
- Access to relevant equipment manual and P&ID drawings.
- CMMS access to create and close work orders.

4. Health, Safety and Environmental Requirements

! WARNING

This SOP must not be carried out without appropriate PPE and, where required, a valid Permit to Work (SOP-SAF-006). If in doubt about safety, STOP and consult the Safety Officer (EMP009 / EMP017).

4.1 Identified Hazards

- High-pressure RO system (up to 12 bar) - pressure release risk
- UV radiation from UV sterilizer lamps - eye and skin damage risk
- Chemical dosing agents (chlorine, antiscalant, acid) - chemical hazard

4.2 Emergency Response

In case of any emergency during this procedure: Press the nearest Emergency Stop, alert personnel in the vicinity, call the Safety Officer and Shift Supervisor. Refer to the Emergency Response Plan (SOP-SAF-004) and site evacuation procedure.

5. Detailed Procedure

5.1 Pre-Task Verification

1. Confirm the relevant equipment is available and no conflicting activities are planned.
2. Check shift handover log for any outstanding issues with this equipment.
3. Verify all required materials, tools, and chemicals are available and correctly labelled.
4. Obtain all required permits (PTW, hot work, confined space as applicable).
5. Brief all involved personnel on this SOP before starting.

5.2 Main Procedure

1. Notify the shift supervisor and next upstream/downstream department of planned shutdown.
2. Reduce production rate to minimum before initiating shutdown.
3. Initiate shutdown sequence from SCADA or local panel as per control configuration.
4. Allow equipment to reach rest before applying isolation.
5. Close all isolation valves (suction, discharge, utility) in the sequence specified in the P&ID.
6. For food-contact equipment: initiate CIP sequence before shutdown if required.
7. Apply LOTO if shutdown is for maintenance purpose - see SOP-SAF-002.
8. Record shutdown time, reason, and condition of equipment in the shift log.